

Automated Tax Form Data Extraction Pipeline Using Google Cloud AI

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Project Overview

Problem

Tax form PDFs require manual reading and data entry, which can be time-consuming, repetitive, and prone to errors, especially as document volume increases.

Objective

To build an automated GCP-based pipeline that extracts, validates, and stores structured customer information from PDF tax forms.

01

Extract Customer Fields
using Google Document AI

02

Parse and validate OCR
results using Python

03

Store structured results in
BigQuery

04

Export reviewed results
into CSV format

05

Route records for manual
review

06

Measure accuracy and
performance

Dataset

Input Dataset

- 481 dummy U.S. tax form PDFs
- 401 clean PDFs and 80 noisy PDFs
- Stored in Google Cloud Storage under raw/

This is a clean, legible scan of a 2010 U.S. tax form (Form 990-E) for W-2 income. The form is for the year 2010 and is for the taxpayer Trevor Duncan, 41435 Hughes Drive, New Teneberg, KS 99528-9775. The form is for the year 2010 and is for the taxpayer Trevor Duncan, 41435 Hughes Drive, New Teneberg, KS 99528-9775. The form is for the year 2010 and is for the taxpayer Trevor Duncan, 41435 Hughes Drive, New Teneberg, KS 99528-9775.

clean

This is a noisy, distorted scan of a 2010 U.S. tax form (Form 990-E) for W-2 income. The form is for the year 2010 and is for the taxpayer Trevor Duncan, 41435 Hughes Drive, New Teneberg, KS 99528-9775. The form is for the year 2010 and is for the taxpayer Trevor Duncan, 41435 Hughes Drive, New Teneberg, KS 99528-9775. The form is for the year 2010 and is for the taxpayer Trevor Duncan, 41435 Hughes Drive, New Teneberg, KS 99528-9775.

with noise

Dataset

REISSUED STATEMENT		Employee's social security number 768-05-9800		Safe, Accurate, FAST! Use  Visit the IRS Website at www.irs.gov/efile	
b Employer identification number 83-1015714		1 Wages, tips, other compensation 168250.2		2 Federal income tax withheld 53860.77	
c Employer's name, address, and ZIP code White PLC Inc 21769 Walker Landing Bakerfurst NJ 06199-9819		3 Social security wages 159372.62	4 Social security tax withheld 12192.01	5 State income tax 5988.48	
d Control number 7480956		6 Medicare wages and tips 206499.26	7 Social security tax 159372.62	8 Medicare tax withheld 206499.26	
e Employer's first name and initial Trevor Duncan 41435 Hughes Drive New Teresaberg KS 99528-9775		9 Advance EIC payment 149	10 Dependent care benefits 164	11 Nonqualified plans 12a See instructions for box 12	
f Employer's address and ZIP code 16 State Employer's state ID number AL 703-50-377		13 Salary 149	14 Other (see enclosed Notice to Employee)	15a See instructions for box 12	
17 State wages, tips, etc. 76475.69		18 State income tax 4802.83	19 Local wages, tips, etc. 124601.64	20 Local income tax 29884.95	21 Locality name John Hancock
22 State wages, tips, etc. 89324.57		23 State income tax 4795.84	24 Local wages, tips, etc. 137013.88	25 Local income tax 24146.13	26 Locality name Gary Loop

Form 942
Copy B-To Be Filed with Employee's FEDERAL Tax Return
This information is being furnished to the Internal Revenue Service.

2010 Department of the Treasury-Internal Revenue Service

Old here. Keep lower portion for your records.

REISSUED STATEMENT		Employee's social security number 768-05-9800		This information is being furnished to the Internal Revenue Service. If you are required to file a tax return, a negligence penalty or other sanction may be imposed on you if no income is taxable and you fail to report it.	
b Employer identification number 83-1015714		1 Wages, tips, other compensation 168250.2		2 Federal income tax withheld 53860.77	
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2010 Safe, accurate, FAST! Use  Department of the Treasury-Internal Revenue Service

clean

REISSUED STATEMENT		Employee's social security number 679-16-9791		Safe, Accurate, FAST! Use  Visit the IRS Website at www.irs.gov/efile	
b Employer identification number 26-7087140		1 Wages, tips, other compensation 238334.21		2 Federal income tax withheld 27649.46	
c Employer's name, address, and ZIP code Brown-Garcia and Sons 36311 Debra Plain Johnsburg CO 53149-8071		3 Social security wages 280329.48	4 Social security tax withheld 21445.21	5 State income tax 5551.41	
d Control number 5480454		6 Medicare wages and tips 280329.48	7 Social security tax 21445.21	8 Medicare tax withheld 191428.01	
e Employer's first name and initial Victoria Morales 12772 Price Highway New Scott AL 92936-1875		9 Advance EIC payment 261	10 Dependent care benefits 163	11 Nonqualified plans 12a See instructions for box 12	
f Employer's address and ZIP code 16 State Employer's state ID number MD 114-53-235		13 Salary 261	14 Other (see enclosed Notice to Employee)	15a See instructions for box 12	
17 State wages, tips, etc. 12326.35		18 State income tax 12326.35	19 Local wages, tips, etc. 226218.83	20 Local income tax 37997.84	21 Locality name Wilde Stone
22 State wages, tips, etc. 186464.21		23 State income tax 119910.52	24 Local wages, tips, etc. 12310.97	25 Local income tax 30427.65	26 Locality name Hocking Negro

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2010 Safe, accurate, FAST! Use  Department of the Treasury-Internal Revenue Service

with noise

Dataset

Evaluation Sample

27

Total PDFs Manually Checked

81

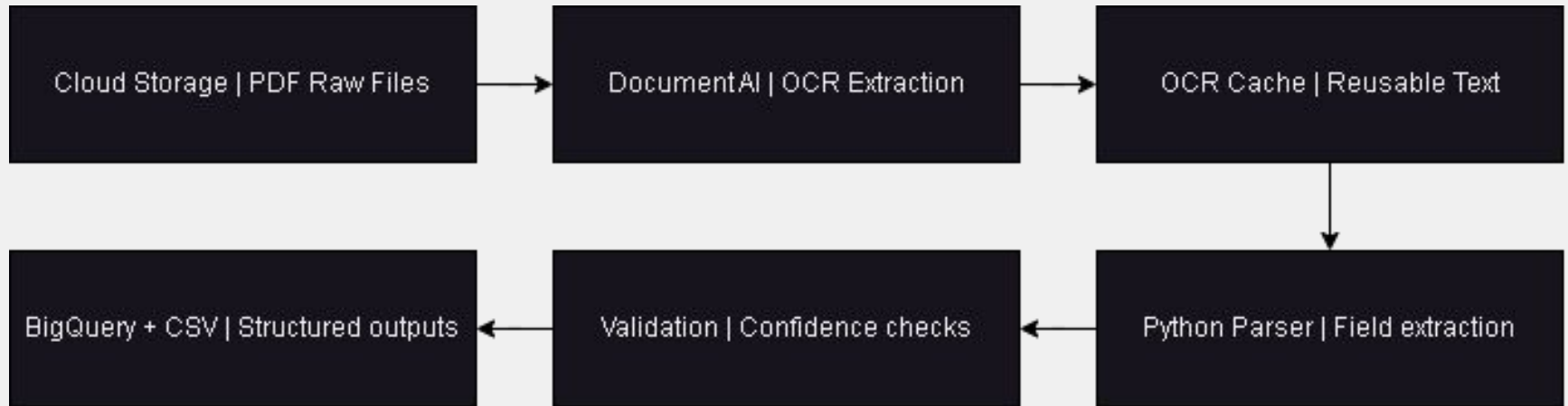
Field-level Checks

11 Clean PDFs | **16 Noisy PDFs**

Note: Noisy PDFs were denoised/enhanced before manual review to make readable fields easier to verify.

▪ **4 Not Verifiable fields**

Pipeline Architecture



Extraction Process

Extraction

Document AI extracts OCR text from each PDF.

Python parser identifies:

- Employee / customer name
- Employee / customer address
- Employee / customer identification number

Form W-2, Wage and Tax Statement, 2010. The form is for Taylor Cox, 613 Roger Crest Apt. 802, Leeton IA 26442-8249. The employer is Carlson Group Group, 56319 Underwood Views, Lake Debramouth NE 33667-9751. The form shows various tax fields including Social Security Number (678-77-709), Gross Income (104567.25), and Federal Income Tax (19328.85).

Box	Field	Value
1	Employee's Social Security Number	678-77-709
2	Employer's Identification Number	940-46-434
3	Wages, tips, other compensation	104567.25
4	Federal income tax	19328.85
5	State income tax	282
6	Local income tax	307
7	Retirement plan	X
8	Health insurance	X
9	Life insurance	X
10	Disability insurance	X
11	Other (see instructions)	
12	Employer's name (do not write)	Carlson Group Group
13	Employer's address (do not write)	56319 Underwood Views Lake Debramouth NE 33667-9751
14	Employee's name (do not write)	Taylor Cox
15	Employee's address (do not write)	613 Roger Crest Apt. 802 Leeton IA 26442-8249

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| Extraction Logic

01

Noise Removal

Cleaning document artifacts for better OCR quality.

02

Regex Extraction

Use regex patterns to detect names, addresses, SSNs, and EINs

03

Scoring Engines

Rank candidate fields using rule-based heuristics

04

Name Re-ordering

Clean formatting and reorder names when needed

05

Auto-approval Logic (Validation)

Apply confidence thresholds and assign final status

Noise Removal

01

Remove Numeric Noise

Filters unrelated money values, box codes, and standalone numbers.

02

Exclude Form Labels

Removes repeated W-2 labels such as “Department of Treasury” and “Internal Revenue Service”.

03

Filter OCR Artifacts

Removes common OCR noise such as e-file, www, safe, and statement.

04

Normalize Text

Collapses extra spaces and removes single-character noise.

Regex Pattern Extraction

01

ID Number Extraction

Detects customer ID format across documents using specialized numeric masks.

###-##-####

02

ZIP Code Detection

Identifies standard 5-digit and extended 9-digit postal codes within address blocks.

or #####-####

03

Street Detection

Finds lines that start with a street number and contain address keywords

Address anchors

Drive, Ave, Plaza, Suite, Apt, Street, Highway

Address linking

Street + city + state + ZIP

NLP Integration (spaCy)

01

Entity Recognition

Differentiates between persons and geographic/corporate entities.

✓ "John Doe" (PERSON)

✗ "Albany" or "Acme Corp"

02

Dual-Pass Strategy

Maximizes name recognition probability by analyzing text variants.

Pass 1:

Original string check

Pass 2:

Swapped version analysis

Dynamic Name Reordering Logic

W-2 OCR can return names in different orders because of field label placement.

OCR Scanned Variant:

Last Name: **Cox**

First Name: **Taylor**

Result: "**Cox Taylor**"

The parser checks nearby label order to determine the correct structural sequence.

If "Last Name" label appears before "First Name", the parser swaps the order.

Reordered Format

Taylor Cox

Extraction Validation Logic

01

Name Validation

- Rejects company suffixes (LLC, Inc., Corp)
- Removes OCR noise & single characters
- Requires at least two valid part names
- Rejects all-uppercase noisy strings

02

Address Validation

- Requires a street number
- Recognized road terms (Street, Ave, Hwy)
- Requires State and ZIP code
- Validates level of detail

03

ID Number Validation

- Standard format: XXX-XX-XXXX
- Prevents merged digit errors
- Rejects missing or incomplete values

Evidence-Based Confidence Scoring

Repeat Evidence

W-2 forms often contain repeated copies of the same fields.

- **Repeated OCR matches** increase confidence scores by verifying data consistency.
- **Missing or invalid** fields receive a baseline 0.00 confidence score.

Confidence Scoring Map

0.00 No match or invalid format

0.90 Valid field found once

0.95 Field confirmed multiple times

AUTO_APPROVED Threshold: ≥ 0.90

Automated Approval Workflow

AUTO_APPROVED

Ready for Structured Output

The record is automatically approved when all required customer fields pass validation.

Requirements:

- ✓ Name confidence $\geq 90\%$
- ✓ Address confidence $\geq 90\%$
- ✓ ID confidence $\geq 90\%$
- ✓ Overall confidence $\geq 90\%$

NEEDS_REVIEW

Sent for Manual Review

The record is routed for review when one or more required fields are incomplete or uncertain.

Common Triggers:

- ✗ Missing fields
- ✗ Invalid ID format
- ✗ Incomplete address
- ✗ Low confidence score

Accuracy and Evaluation Results

Field-Level Accuracy

88.31%

Clean PDF Accuracy

87.88%

Noisy PDF Accuracy

88.64%

Auto-Approval
Precision

100.00%

Evaluation was based on 27 manually checked PDFs with 81 field-level checks. Fields marked as Not Verifiable were excluded from the accuracy calculation.

Accuracy = Pass / (Pass + Fail)

68 / (68 + 9) = 88.31%

| Full Dataset Processing Summary

Total PDFs Processed

481 / 481

Auto-Approved

258
53.64%

Needs Review

223
46.36%

Processing Performance

Avg. Processing Time:

19.56s / PDF

Total Processing Time:

~ 2.61 hours

(9,408.36 total seconds)

Dataset Composition

Clean PDFs **401**

Noisy PDFs **80**

Business Impact

Operational Efficiency

The pipeline reduces the manual workload by automatically ingesting and encoding tax form PDFs.

Data Readiness

Outputs are delivered as structured CSVs and BigQuery tables, enabling immediate BI reporting.

Pipeline Processing Summary

258

Auto-approved (53.64%)

223

Manual Review (46.36%)

Volume Handled

Successfully processed 481 of 481 PDFs. Uncertain records are routed based on confidence levels.

The system is designed to maximize output via automation while maintaining high processing reliability.

Limitations & Future Vision

CURRENT LIMITATIONS

- Very noisy, rotated, or cut-off PDFs may still need manual review
- OCR quality affects extraction accuracy
- Address extraction is harder because it requires street, city, state, and ZIP
- Rule-based parsing may not handle every layout variation
- Ground truth sample is limited
- Validation is intentionally strict to avoid false auto-approval
- Evaluation only covers three required customer fields

FUTURE IMPROVEMENTS

- Add layout-aware extraction for better field positioning
- Add human-in-the-loop review dashboard
- Improve noisy PDF handling with visual fallback models
- Integrate approved records directly into a CRM workflow
- Add monitoring dashboard for status and accuracy
- Extend validation to employer fields (EIN, Name, Address)
- Improve auto-approval rate while maintaining data quality
- Reduce processing time

The system is designed to automate reliable records while safely routing uncertain records for review.